COMP1830 BLOCKCHAIN FOR FINTECH APPLICATION

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1. Introduction

Know Your Customer (KYC) is an essential practice that is carried out in the financial sector, acting as a foundational feature that adheres to regulations and legal requirements to help mitigate fraudulent activities for both new and existing client explains (Yadav & Chandak, 2019). It is further emphasized that conventional KYC methods rely on collecting and verifying customer data via centralised systems which frequently encounter challenges such as inefficiency and increased regulatory costs. As reported by (Deloitte, 2018), financial organizations can spend as much as $60 million annually, supporting that statement (Reuters, 2016) also reported that some organisations can spend up to $500 million to comply to KYC and Customer Due Diligence (CDD).

Moreover, this conventional approach presents considerable amounts of treats to data privacy and security. These centralized databases are particularly vulnerable to cyberattacks, which is evident by the data breach of Wonga in 2017, where over 270,000 customer records were compromised (Chin, 2024). Cyber attacks like these highlights the necessary need to find a secure and efficient alternative.

The blockchain technologies presents itself as a possible solution to these issues. (Hussain & Usmani, 2019) expresses, that this will help establish a unique framework that utilizes a self-sovereign and decentralised KYC method. This framework will aim to improve customer privacy by implementing consent-driven mechanisms. It will also incorporate regulatory oversight and assist financial organisations by leveraging reliable and precise customer information, all while minimizing the cost associated with customer acquisition.

This report will investigate the potential of blockchain technologies transforming identity verification and customer onboarding. Current methodologies will be examined to suggest a blockchain driven approach and assess any possible enhancements using both qualitative and quantitative analytics. Moreover, this report will include a critical evaluation and discuss prospects for this adoption with in the financial sector.

2. Traditional Practices In KYC Processes

The Traditional KYC processes played a vital role in protecting companies from operating businesses with individuals or organizations who are involved in money laundering, terrorism financing and corruption (Swift, n.d.). The financial institutions have to verify the client's identity, evaluate all risks as well as oversee all transaction activities. This approach relies on a centralized data gathering and storing by using manual verification processes which poses various challenges.

2.1 Challenges In Traditional KYC Processes

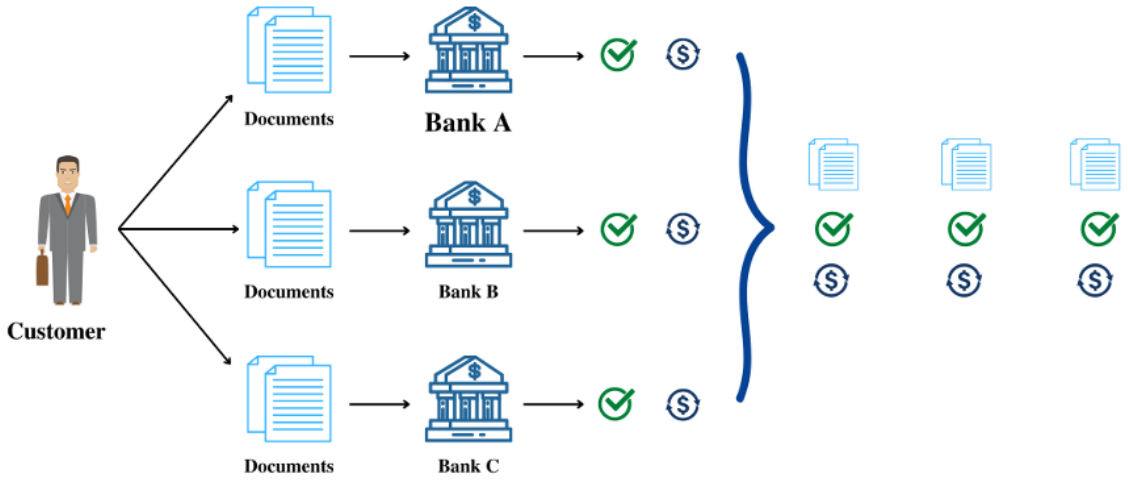


Figure 1 Traditional KYC Practices (Mansoor et al., 2023)

Figure 1 Traditional KYC Practices Figure 1 Traditional KYC Practice represents how the current KYC compliance is carried out in the financial sector. This approach is characterised as time consuming and high maintenance because the customer is required to undergo the verification process multiple times for every organisation they interact with. Furthermore, customer data is verified with a wide array of documentation including passports, utility bills and bank statements which delays the whole process leading to poor customer experience (Mansoor et al., 2023).

As reported by (Reuters, 2016) and (Deloitte, 2018), adhering to KYC compliance is substantially expensive with organisation spending millions annually. The manual verification together with third party services inflate the cost even further. Even though businesses employ adequate data security for collecting and storing sensitive customer data, even one hack can lead to large amounts of data to be stolen which can lead to identity theft and other types of cybercrime. Cross border operations means that financial institutes will have to deal with different KYC requirements in different jurisdictions which can lead complexities. To ensure that the KYC process is adhering to these requirements, these financial institutes will have to implement cumbersome procedures to maintain consistency which add to the challenge and complexity to an already complicated process that leads to higher costs and increase the time taken to complete the KYC identity verification process (Fraud.com, n.d.).

2.2 Industry Example: Starling Bank

The recent challenges experienced by the Starling Bank underscores the inefficiencies and obstacles with KYC procedures. Starling had failed its regulatory compliance in 2023 with their KYC and AML measures. Starling had developed an automated screening protocol for onboarding customers due to an increase in customer base. This screening failed as it was only screening a fraction of the full list of those customers subjected to financial sanctions (ICA, 2024). They developed this to reduce cost and the time taken to carry out the KYC procedure. Whereby, this in turn attributed to their failure due to lack of manual supervision. This situation highlights how centralized and fully automated frameworks can fail, which many other financial institutions face regularly in the market place.

3. High-Level Architecture Solutions

To solve the traditional KYC problems, by making use of the blockchain technology's decentralised, immutability and transparency. As the solution is based on the idea of self-sovereign identity, which will allow individuals to maintain complete authority over their own personal data while also being able to select which information they can reveal to these financial institutions. This will allow to manage time and cost by reducing multiple verification processes that each individual has to go through as it is already completed.

This process has been identified by going through the research findings of (Hussain & Usmani, 2019), (Yadav & Chandak, 2019) and (Mansoor et al., 2023).

The procedure will start with a reliable organisation that will verify the user's identity and subsequently issue the records as encrypted tokens which will be securely stored into their digital wallet. These tokens will than be recorded only as hashes onto the blockchain. When the financial institutes wan to initiate the KYC process, they will request it through smart contracts. The users will retain the authority to either grant or refuse access according to their preferences. All this while, the entire process will permanently be recorded onto the blockchain which will allow both transparency and ability to conduct audit checks for AML.

The possible technologies leverages in this solution are to make use of Hyperledger Fabric. (Ullah et al., 2021) describes Hyperledger fabric as a private blockchain network that only authorized users can exchange information. It allows to establish channels among groups to share distinct records exclusively among designated members. The key features that they offer are Identity Management, Security and Privacy, Chaincode Functionality and Modular Architecture. It utilizes smart contracts like Chaincode to manage their assets and transactions. These immutable transactions are than recorded in the blockchain ledger and preserved within their database. (Ullah et al., 2021) research concluded by explaining that the Hyperledger Fabric can safely and privately store the customers KYC documentation and guarantees it being time saving, reduced costs and bring transparency in traditional KYC practices.

As the aim is to create a new KYC system which is self-sovereign, by using Zero Knowledge Proofs (ZKPs) will be essential. As (Pauwels, 2021) research explains that ZKPs approach in KYC will avoid sharing all type of personal information with every organisation the user interacts with. This will also allow the user to be identified legally while also allowing the user to gain their privacy without compromising the required transparency.

4. Analysis Of Improvement

4.1 Qualitative Benefits

There are many qualitative benefits of this blockchain based KYC solution. As (Malhotra et al., 2022) describes that as the blockchain is immutable, secure and anonymous which will reduce the on boarding time, risks, costs and improves overall customer experience. Moreover, blockchain technology removes any single point of failure and large-scale data breaches that occur in centralised databases due to its decentralised data storage. Jose Parra additionally describes the use of self-sovereign identity as all the information stored would be owned by the customers and not these financial institutions which will cause a paradigm shift which will complement the GDPR consensus of allowing users to know and manage their own data. This approach will align with the privacy

regulations that are set by the GDPR and build adequate trust between the customers and financial institutes with the transparency that the blockchain offers.

4.2 Quantitative Benefits

The KPMG group worked in Singapore with financial institutions and regulatory authority to develop a similar KYC utility on the blockchain platform. Their prototype resulted in saving 20-25 percent of the cost by reducing duplication and providing a clear audit track while also offering stability, efficiency and security (KPMG, 2018). Their utility helped by reducing the onboarding process and eliminate the need to repeat the KYC process multiple times with each different vendor.

4.3 Design Decision

The design choices made in the solution, include selecting a blockchain platform and a consensus protocol which will have significant impact on the efficiency and scalability. By making use of permissioned blockchain like the Hyperledger Fabric that has robust identity features enhance the efficiency of KYC identification. Furthermore, it addresses the to the inefficiency that cone with redundant execution of similar verification and facilitating secure data sharing, while being cost effective as well as transparent (Ullah et al., 2021). The implementation of ZKPs guarantee that sensitive data are confidential during the verification process while also mitigating any privacy concerns. This technique challenges the traditional distinction between privacy and transparency, providing a framework that is transparent and beneficial to all stakeholders (Pauwels, 2021).

5. Critical Evaluation

5.1 Benefits Of Proposed Solution

By decentralising the data storage there will be no single organisation that will possess total authority over the data eliminating the single point failure found in centralised databases. There is no requirement for independent verifications, individual data collection or mange separate verification processes which will remove the redundant verifications and long onboarding procedures leading to reduced operational cost and improved efficiency. Furthermore, by using ZKPs, all sensitive information is shared on a need-to-know basis which are permanently encoded on to the blockchain which will make It harder to make changes to the data. Additionally, the blockchain will allow to make changes to evolving regulations set by the GDPR (Singh, 2024).

5.2 Challenges And Limitations Of Proposed Solution

It can get expensive to implement this solution in the initial stage, especially smaller financial institutes as they have limited funds to adopt new technologies. Conversely larger financial institutions have their own legacy centralised frameworks which will conflict with the decentralised agenda whereby requiring substantial research which is costly and time consuming. Even though ZKPs mitigate privacy concerns with disclosing sensitive data, it is still considered a developing technology and still requires further advancement to achieve scalability. In all, blockchain is still considered a new technology and people are still unsure and unfamiliar with how the system will work and resist adoption (Singh, 2024). There are also chances that this new technology will be expensive to run because the blockchain servers will constantly have to be mined to keep the vast financial market afloat. This in turn could add to the global warming effect that the blockchain system already adds, plus this increased data on the entire financial market could burden it.

5.3 Personal Evaluation

The proposed solution marks a substantial improvement in mitigating the shortcomings associated with traditional KYC practices. However, its effectiveness will depend on the industry wide acceptance and compliance with regulatory bodies like GDPR. If financial institutions, regulatory bodies and developers do not cooperate, the proposed advantages of the blockchain based KYC may be restricted to specific applications.

6. Conclusion

After analysis the current Know Your Customer (KYC) practices, revealed that there are significant inefficiencies and vulnerabilities such as exponential compliance cost, multiple verification procedures causing redundant processes and considerable risks to data security which correlates to decreases the customer experience. To solve this problem blockchain technology has emerged to combat these issues by its decentralised, transparent and immutable framework.

The report proposes a solution by making use of self-sovereign identities. This process starts with users verifying their identities with some sort of governing bodies that will save the users encrypted documents into digital wallets. By utilizing Hyperledger Fabric, financial institutions can issue smart contracts to their customers, where the customer has to accept which information they want to reveal. Together with the use of ZKPs consensus protocol, it will allow confidentiality during the verification process while maintaining full transparency.

The proposed solutions offer several key benefits like improving the data security through decentralised storage while also reducing the onboarding process and reduce notable cost that financial institutions incur on a day-to-day basis. By incorporating the ZKPs, it will guarantee data privacy while ensuring scalability compliance with regulatory bodies like the GDPR. Furthermore, the inherent immutability as well as transparency of the blockchain technology will build trust among relevant stakeholders ensuring a strong foundation is built for these new implementations on the KYC practices.

Nonetheless, with every strength comes limitations. Big financial institutions have their own existing legacy systems which they have been using for a long time and have gotten comfortable with it, meaning that to implement this new found practice will require meticulous planning the development process which will incur a lot of cost. Similarly, small financial institutions will not have sufficient funds to develop and implement these technologies. Additionally, fostering an industry wide adaptation will require the whole market place to create a collaborative network to fully leverage and utilize the advantages that this practice proposes. All the while this new technology will be discouraged by people as they are still not aware of what blockchain is and how it is used. This will add to people not wanting to use this practice as they are unfamiliar with process causing people not to adapt this practice. Also, blockchain could greatly contribute to global warming which is already frown upon.

To conclude this report, it was found that blockchain can offer a strong compelling solution to the long-standing problem caused by the traditional Know Your Customer (KYC) practices. Although implementing it will be a complex procedure, the prospects it offers are immaculate as it enhances the data security, lower costs and improves the user experiences. This positions the blockchain technology as a vital advancement for any of the future KYC practices. With the way the technologies are advancing which can be seen within the blockchain industry showcases how influential it will be

and can potentially be benchmarked an a secure, low-cost, efficient and transparent procedure for customer onboarding and the identification verification.

7. References

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Appendix – Lab Tasks

A zip file is also attached of the remix task as you have asked.

Lab Exercise 1.2.1

Hashes of two strings provided in the lab exercise.

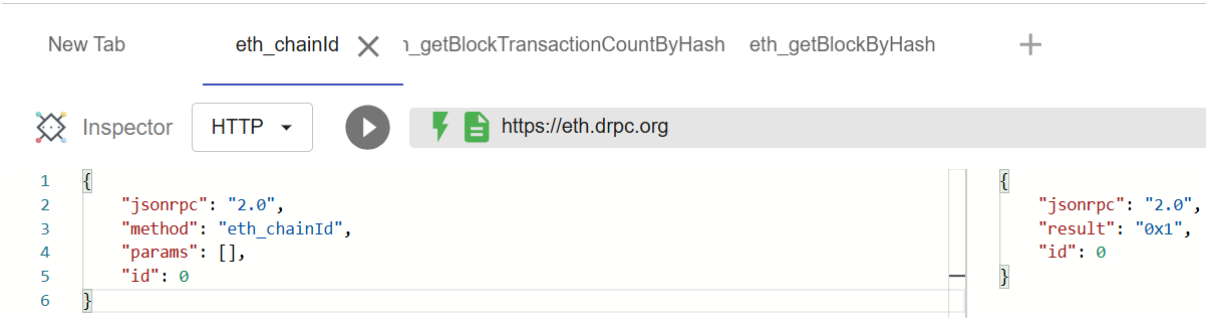
The sha256 hash of the text “COMP1830-Blockchain for Fintech” is
52c7ab834474b736205fb3b3c49421dffb2b5d03ffc203c3923315a3b72f2ae1

The sha256 hash for the text “COMP1830-Blockchain for Fintech!” is
36d80f1d3f8f096dd3975d784f1dd7117cda983ee60580389a988d4f81a26101

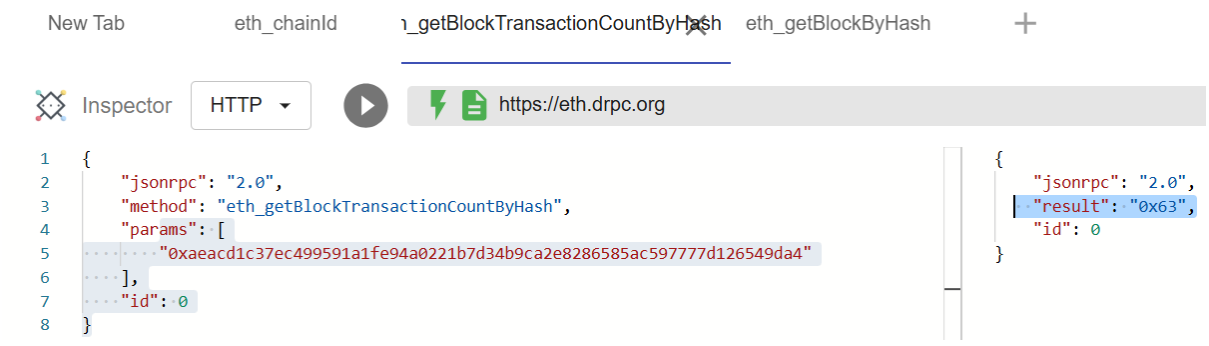
Lab Exercise 2.2.1

Answers to the questions in the lab exercise & screenshot of RCP server response

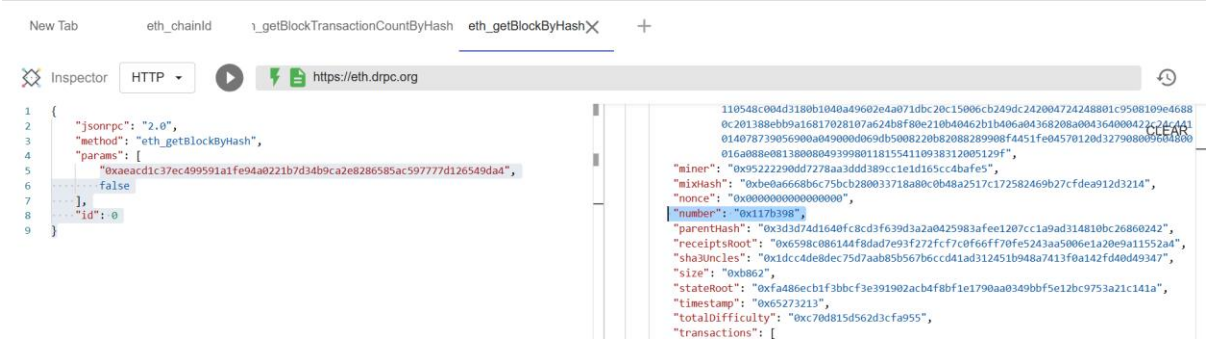
The chainId for Ethereum’s main network is **0x1 (Hexadecimals) or 1 (Decimal)**



There are **0x63 (Hexadecimals) or 99 (Decimals)** transactions

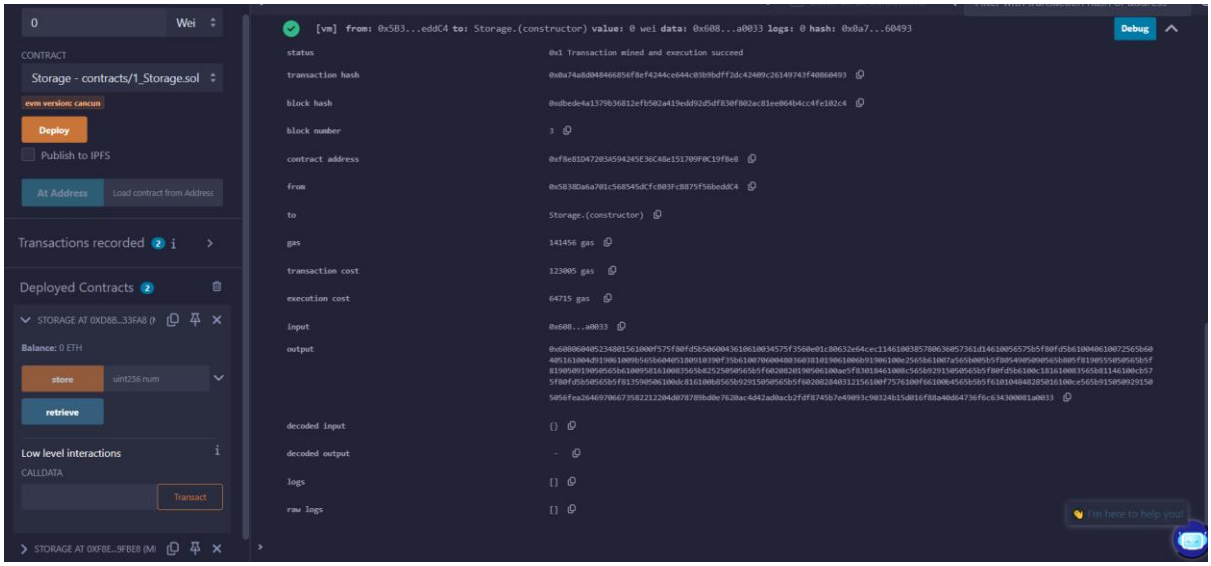


The block number is on **0x117b398 (Hexadecimals) or 18330520 (Decimals)**



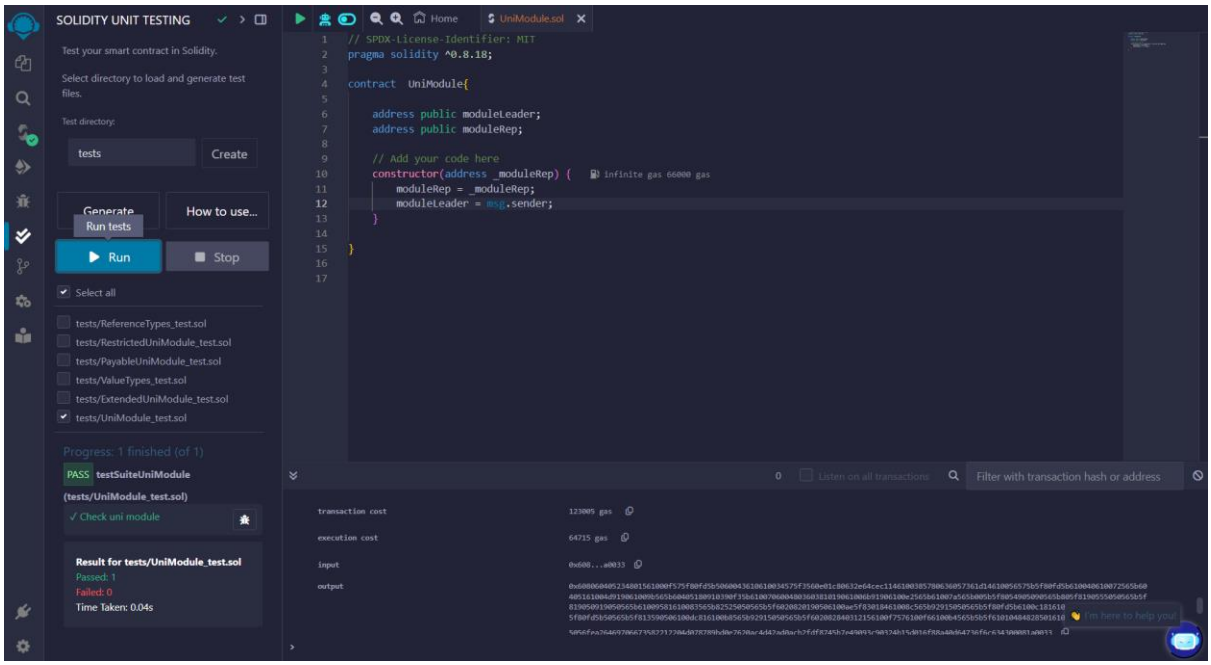
Lab Exercise 2.3.2

Screenshot of contract deployment showing the transaction in Remix



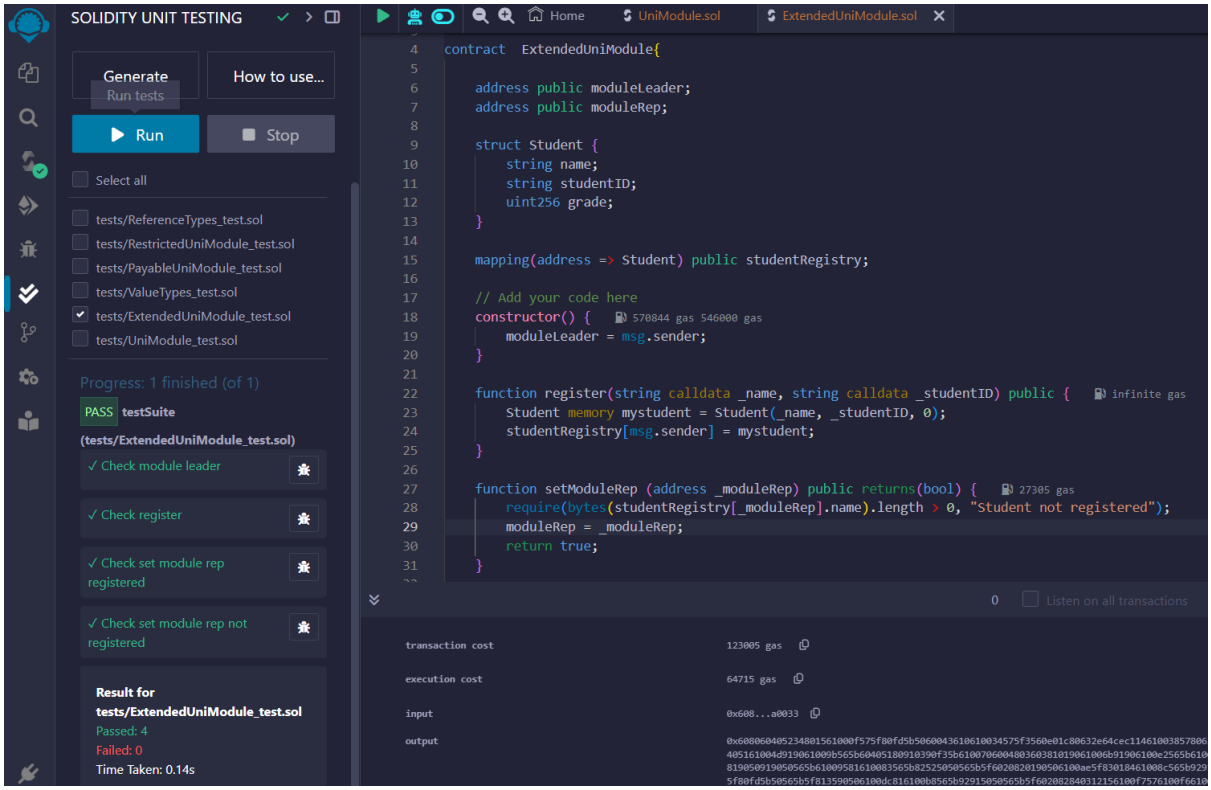
Lab Exercise 3.2.1

Smart Contract Implementation & Screenshots of passing tests



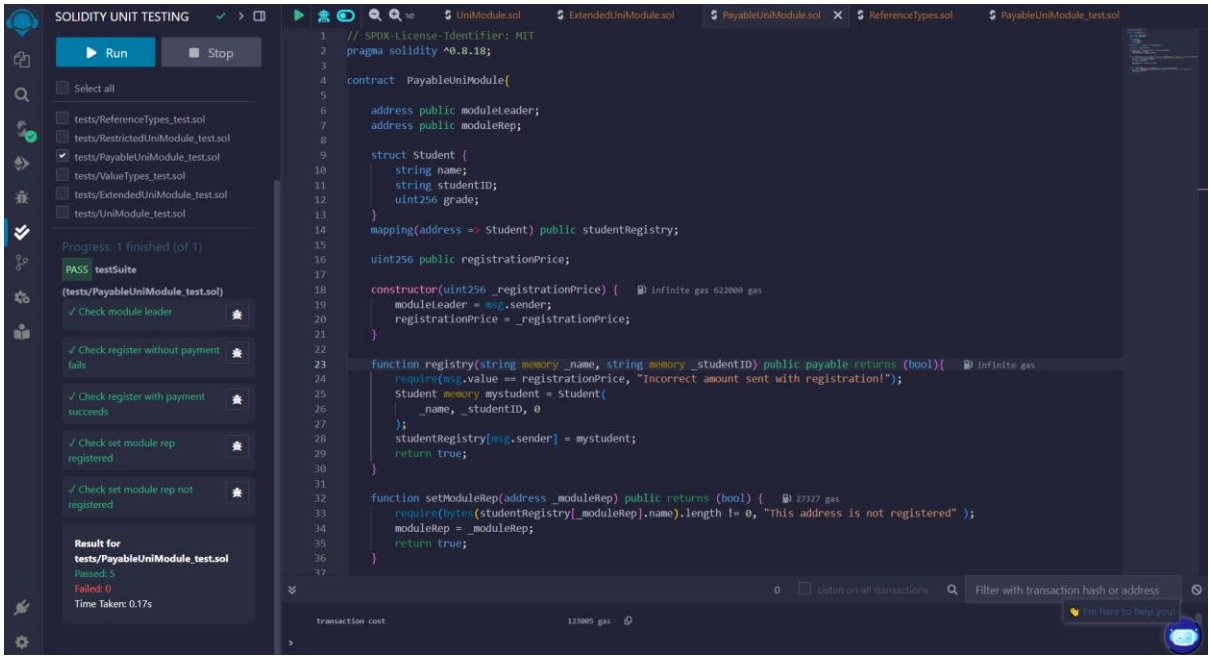
Lab Exercise 3.3.1

Smart Contract Implementation & Screenshots of passing tests



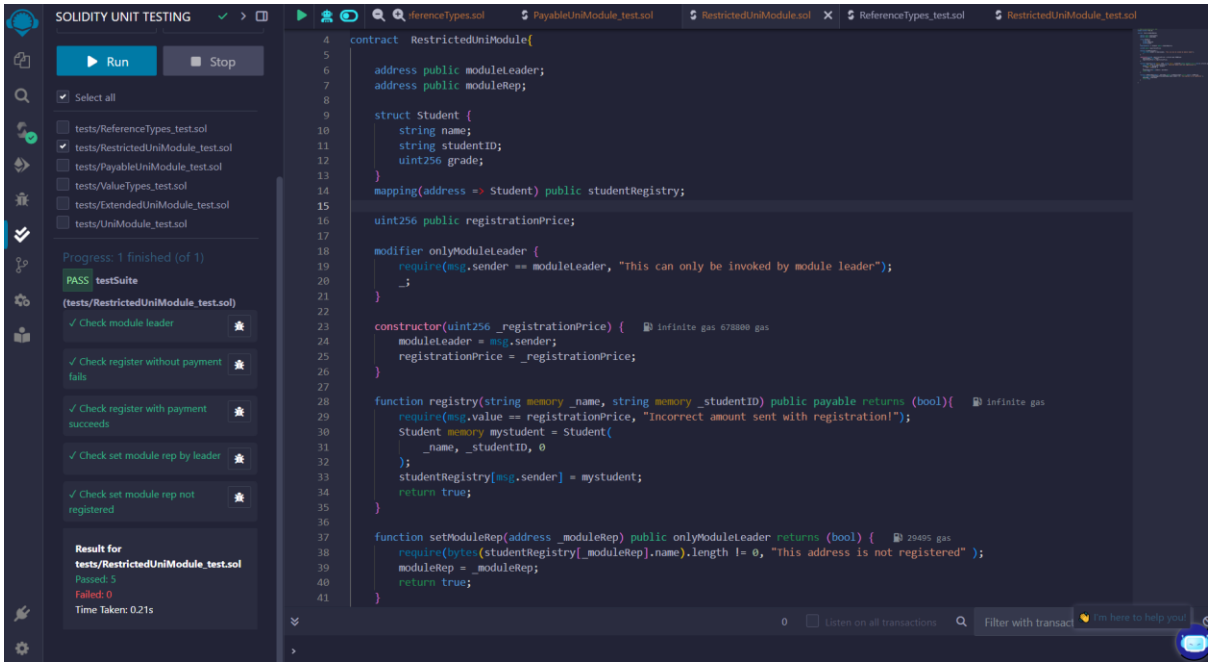
Lab Exercise 3.3.2

Smart Contract Implementation & Screenshots of passing tests



Lab Exercise 3.4.1

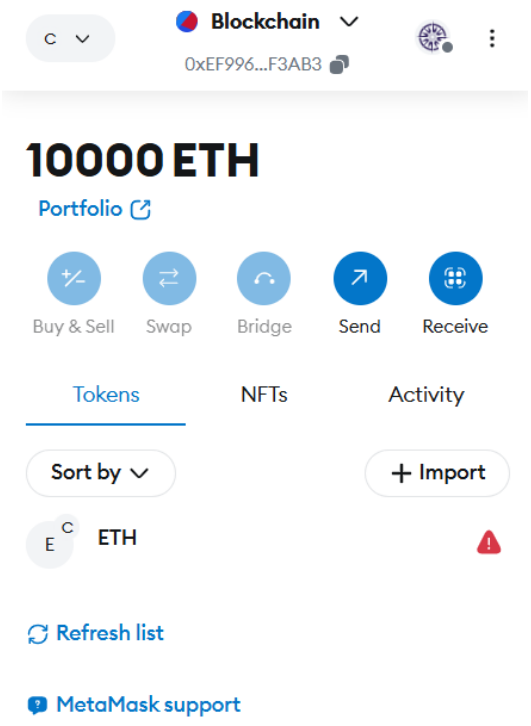
Smart Contract Implementation & Screenshots of passing tests



Lab Exercise 4.2.1

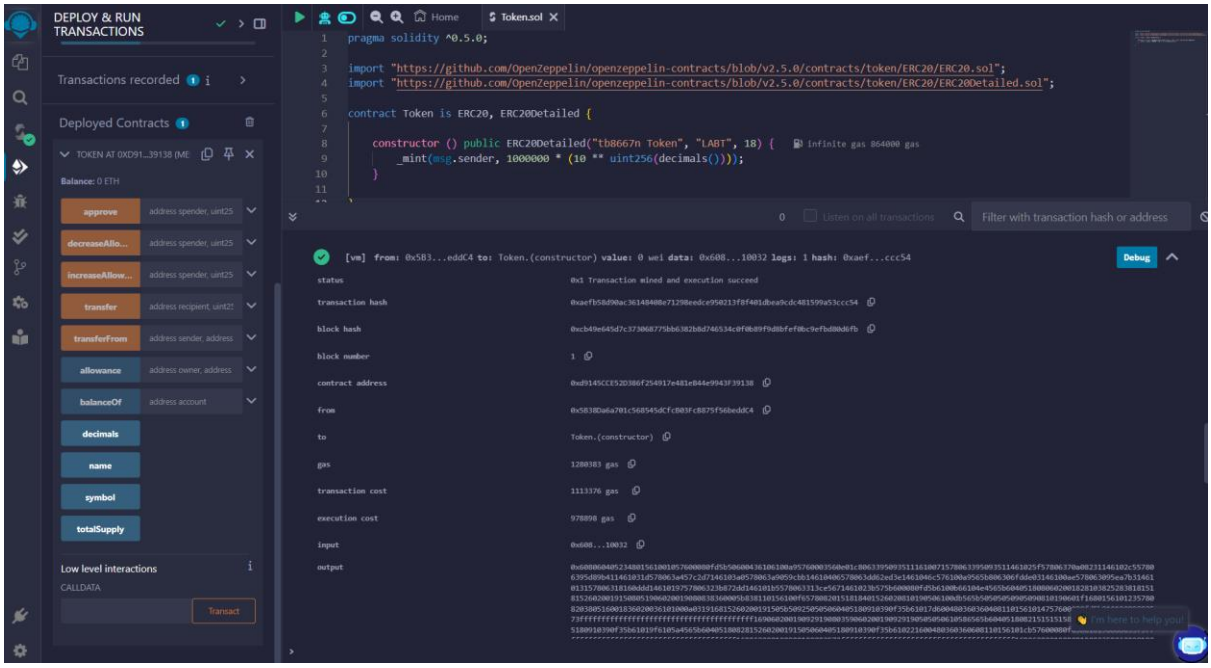
Screenshot of funded account and RPC URL of your tenderly fork

https://virtual.mainnet.rpc.tenderly.co/93ada59e-e1bc-489e-8d7e-74c4ca119abd



Lab Exercise 4.3.1

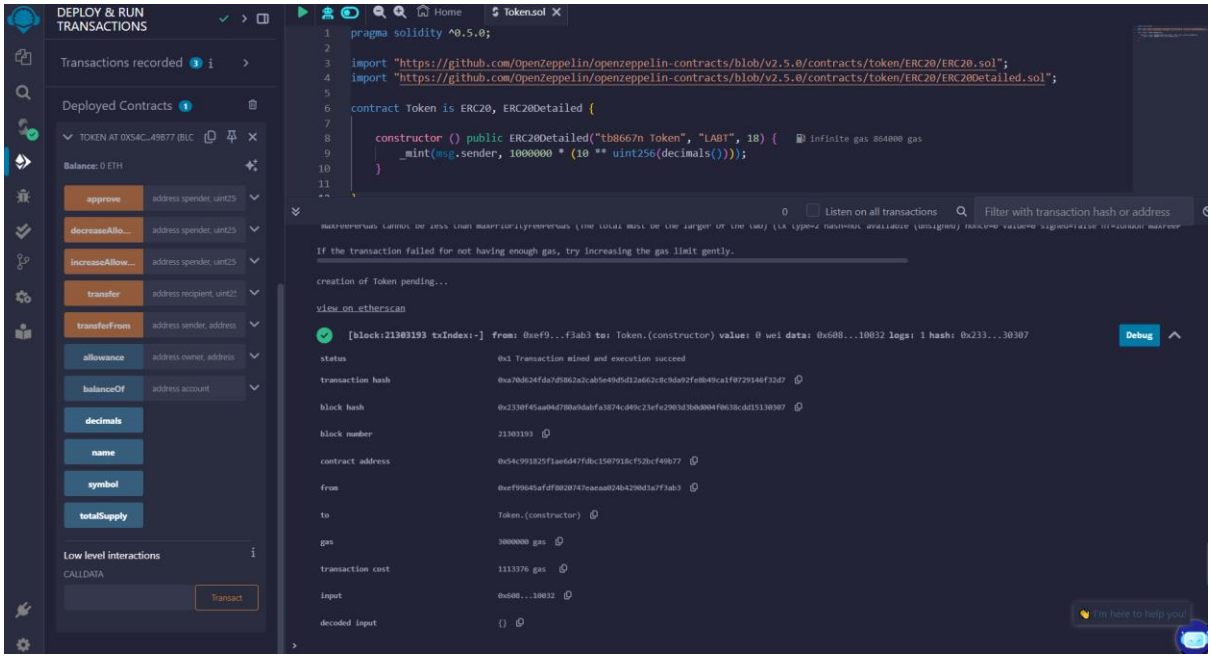
Screenshot of deployment

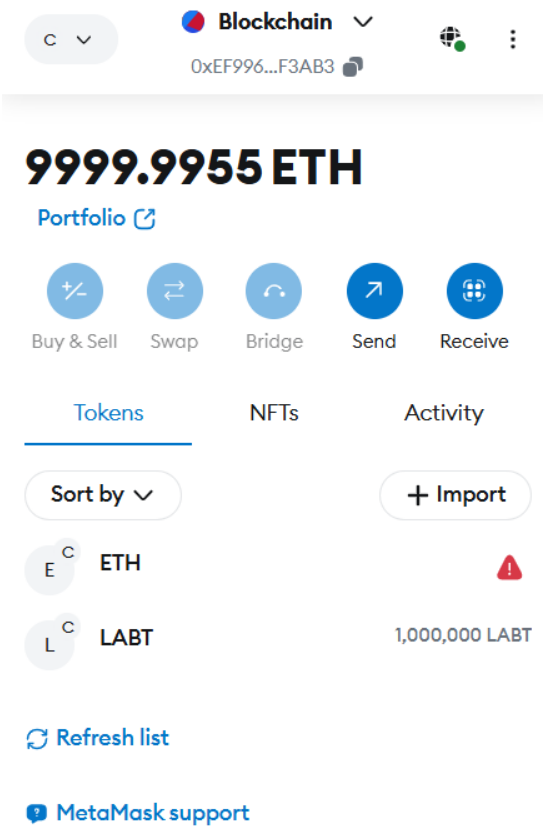


Lab Exercise 4.3.2

Screenshot of deployment and MetaMask showing custom tokens in wallet

Token Address: 0x54C991825f1aE6d47fDbC1507918cF52bCF49b77





Lab Exercise 6.3.1

Initial Quantity in the Pool.

$$x = 10$$

$$y = 30$$

$$k = 300$$

The trader buys 10 Y.

$$30 - 10 = 20$$

To keep the same constant of 300 the pool will re-adjust.

$$x.y = 300$$

$$x.20 = 300$$

$$x = 300/20$$

$$x = 15$$

Which means, after the trade the pool had 15x assets. However, before the trade the pool already had 10x assets therefore the extra 5x is the adjustment from when the trader bought 10 Y assets.

Therefore, the price of 10 Y assets is **5X** because each Y asset is 0.5X and the liquidity pool after the trade will be **15X and 20Y**

Lab Exercise 6.3.2

Initial Quantity in the Pool.

$$x = 1000000$$

$$y = 3000000$$

The constant will be:

$$k = 1000000 * 3000000 = 3000000000000$$

$$3000000 - 10 = 2999990$$

$$x.y = 300$$

$$x.2999990 = 3000000000000$$

$$x = 3000000000000/2999990$$

$$x = 1{,}000{,}003.333344444$$

$$1{,}000{,}003.333344444 - 1000000 = \ 3.333344444$$

The price of 10Y assets is **3.333344444X assets**

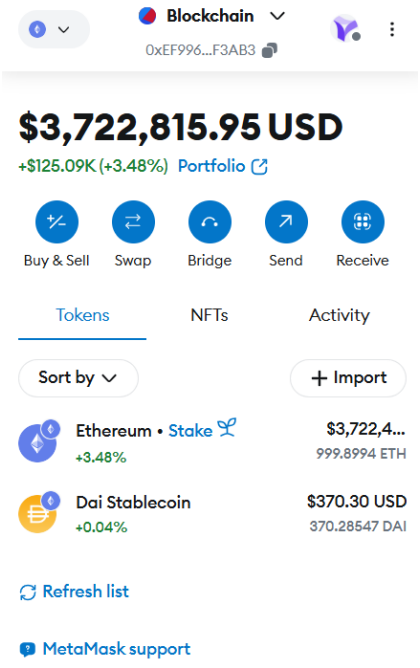
$$\frac{3.333344444}{10} = 0.3333344444$$

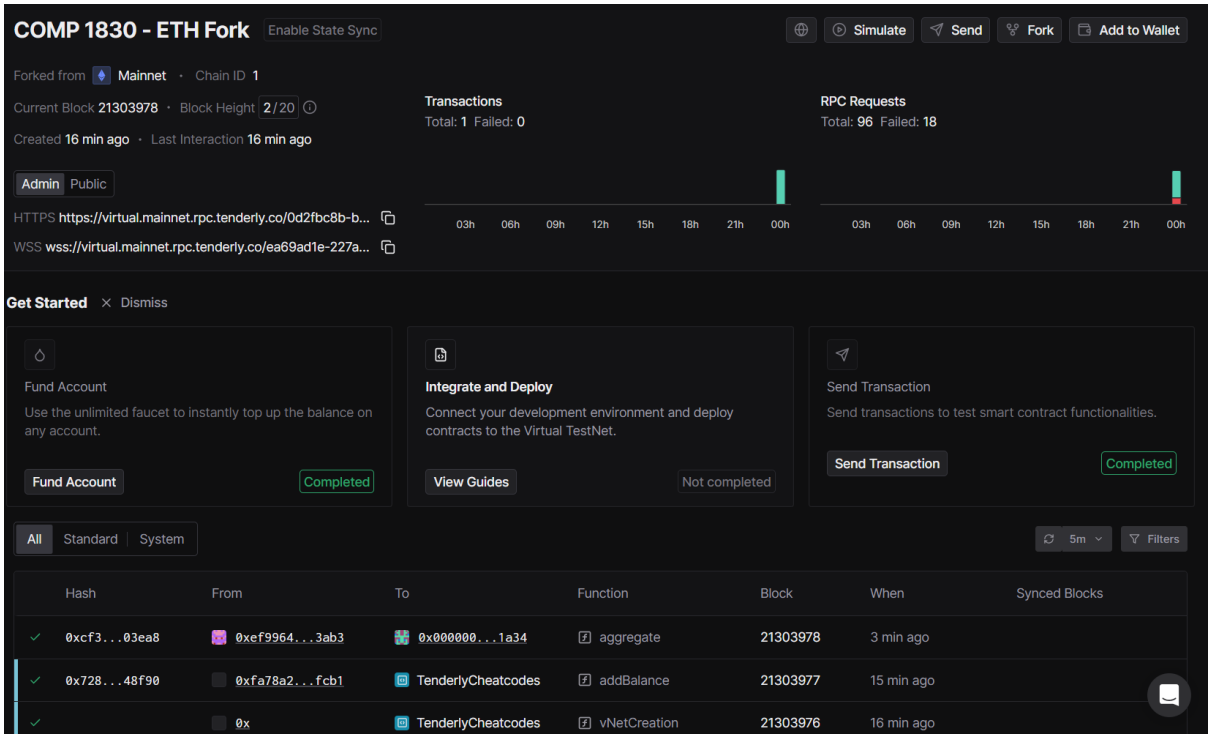
Therefore each Y asset is **0.3333344444X asset**

Lab Exercise 6.3.3

https://virtual.mainnet.rpc.tenderly.co/0d2fbc8b-be0a-47b6-81db-20522226c506

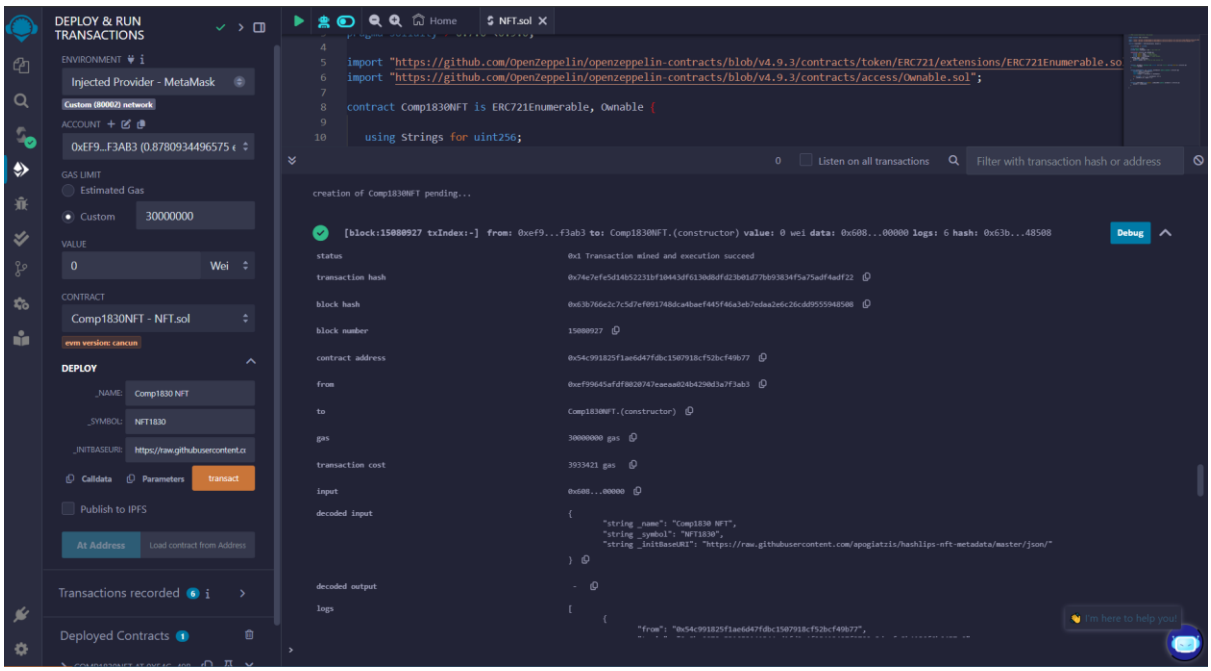
wallet address: 0xEF99645AFdf8020747eaeaA024B4290D3a7F3AB3

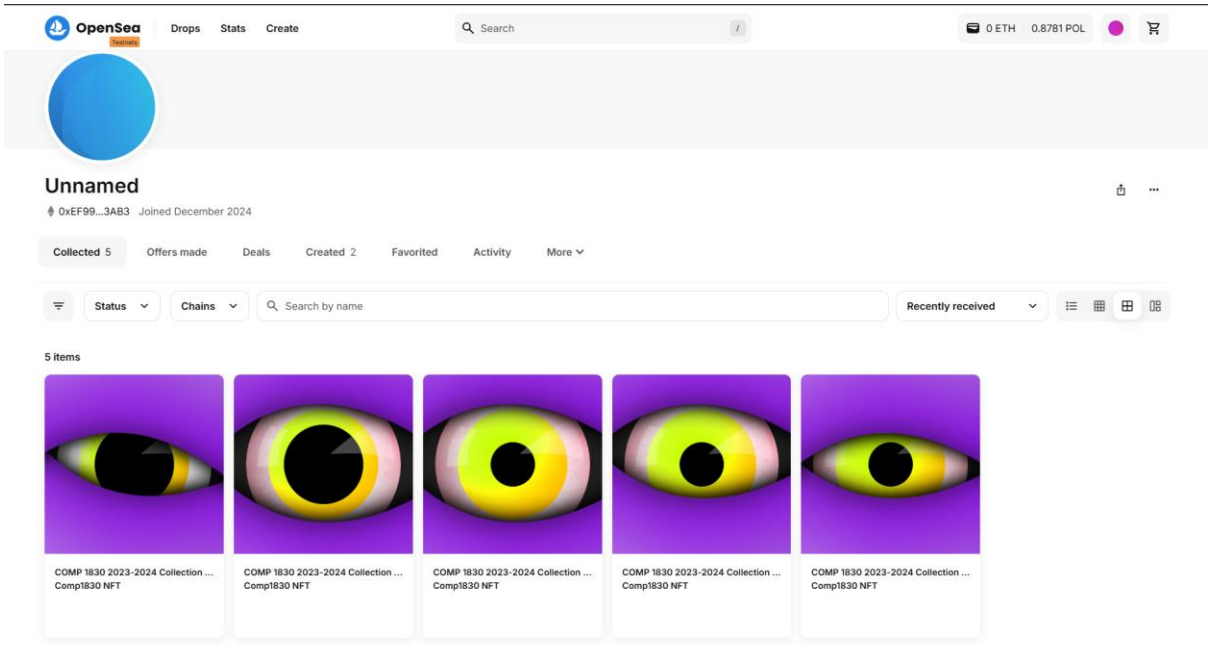




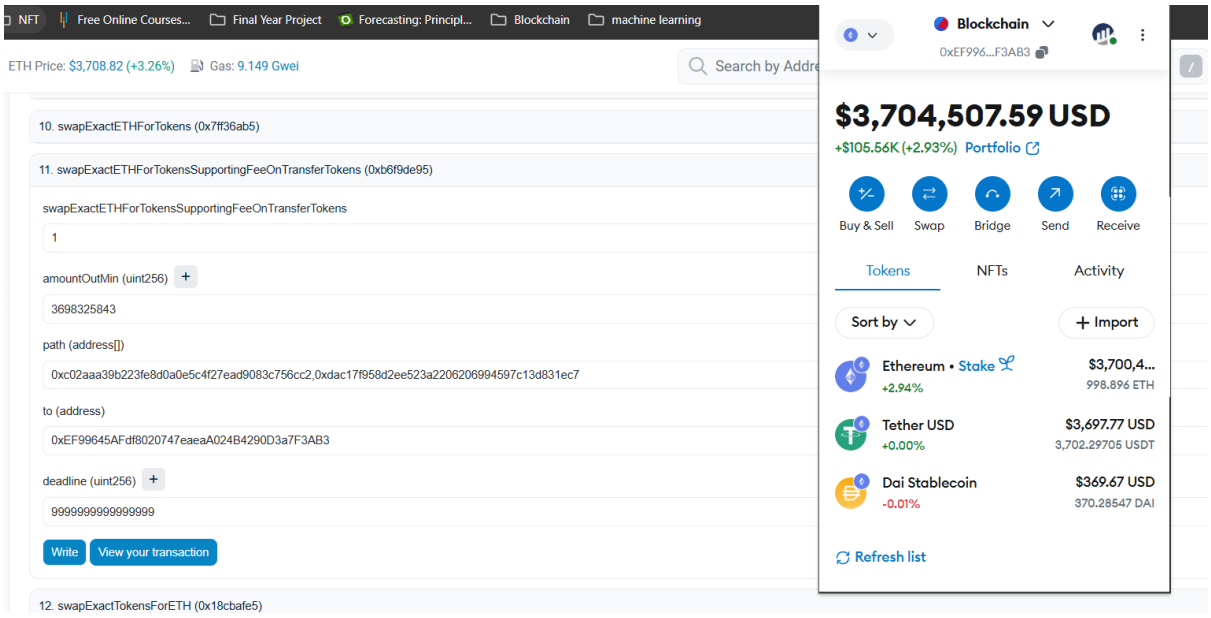
Lab Exercise 6.4.1

NFT Collection Address 0x54C991825f1aE6d47fDbC1507918cF52bCF49b77





Lab Exercise 7.1.1



Lab Exercise 7.3.1

Alice deposited 1 ETH and 100 DAI

She owns 10% (0.1) of the pool

xReservesBefore = 10 ETH
yReservesBefore = 1000 DAI

xReservesAfter = sqrt((10*1000) * (1/400))
xReservesAfter = 5

yReservesAfter = sqrt((10*1000) * (400/1))
yResrvesAfter = 2000

As Alice owns 0.1 of the liquidity, she will get at the time of the withdrawal a total of:
0.5 ETH = $200
200 DAI = $200
Alice got $400

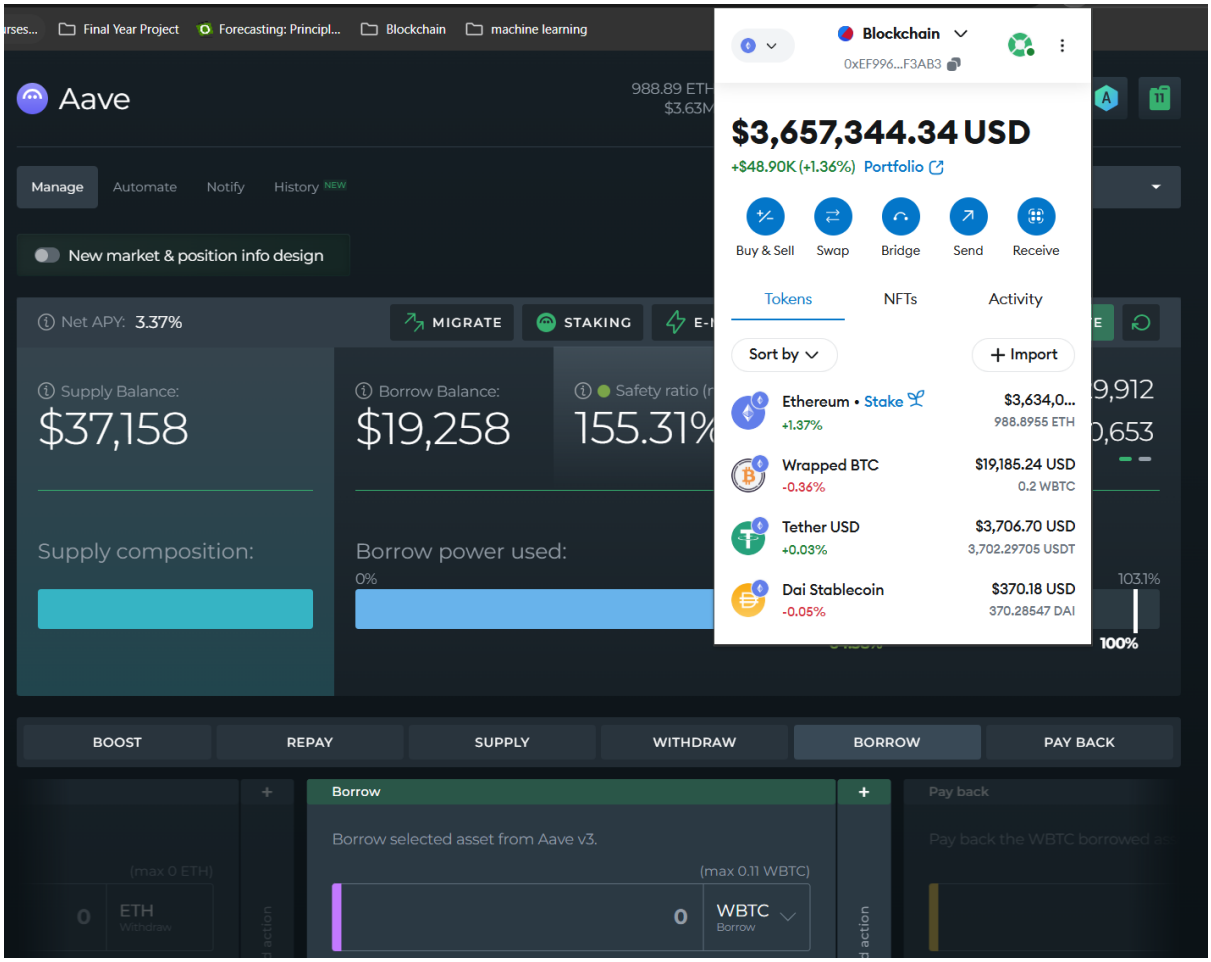
1 ETH = $400
100 DAI = $100

Which is $500 in total

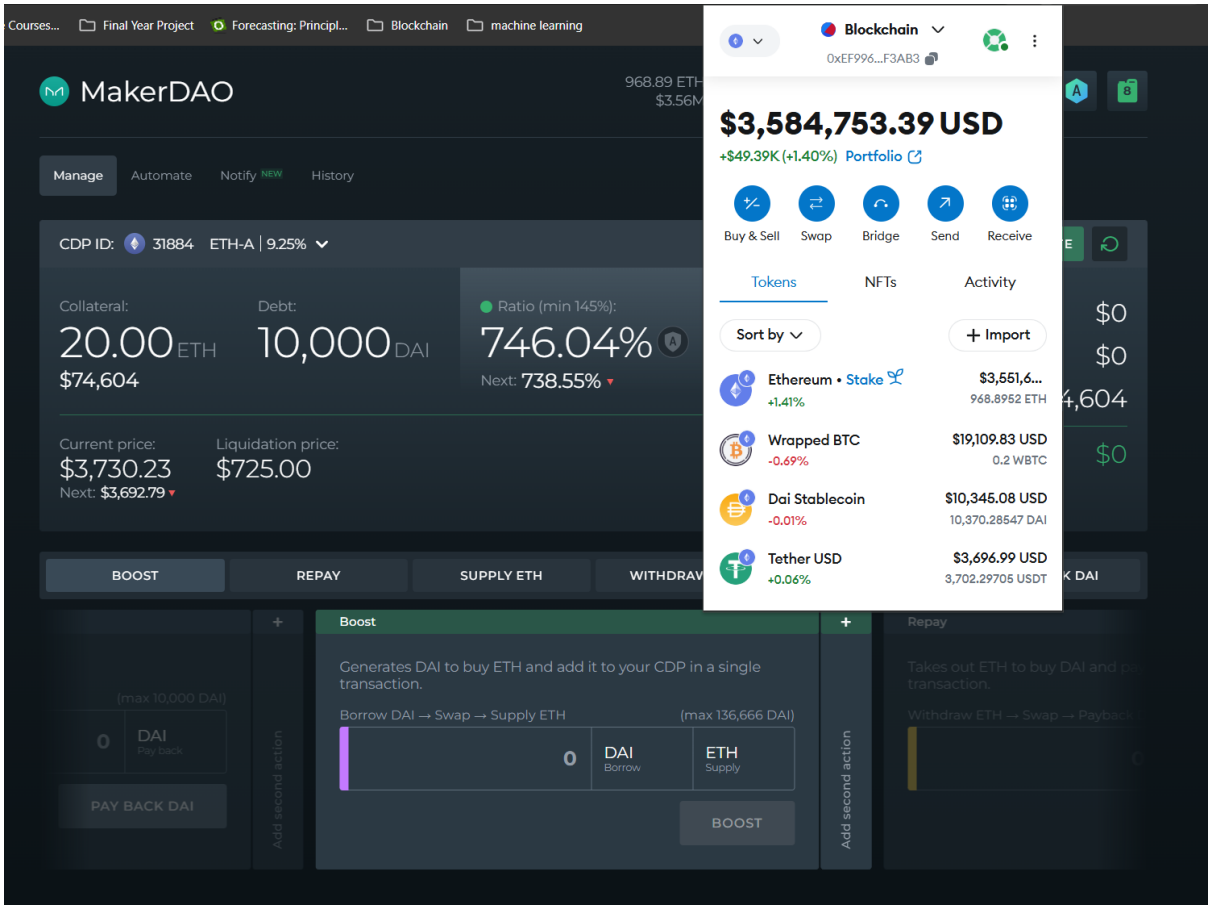
The Impermanent Loss = 400 – 500 = -$100

Alice has impermanent loss of $100

Lab Exercise 7.4.1



Lab Exercise 8.1.2



Lab Exercise 8.2.1

Question 1

What mechanism did Terra's UST use to maintain its peg to the US dollar?

B. An algorithmic process involving another cryptocurrency, LUNA.

Question 2

How did the mechanism to stabilize UST's value work under normal conditions?

C. By allowing UST to be exchanged for LUNA at a stable rate.

Question 3

What was a primary factor that led to the collapse of UST's peg to the dollar?

B. Loss of market confidence and a subsequent sell-off.

Question 4

What is the term used to describe the negative feedback loop that accelerated the collapse of UST and LUNA's value?

C. Death spiral.

Question 5

Which of the following is a key difference between UST and other stablecoins like USDT (Tether)?

B. UST's value is maintained algorithmically and is dependent on another cryptocurrency's value.